

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

Data Preview

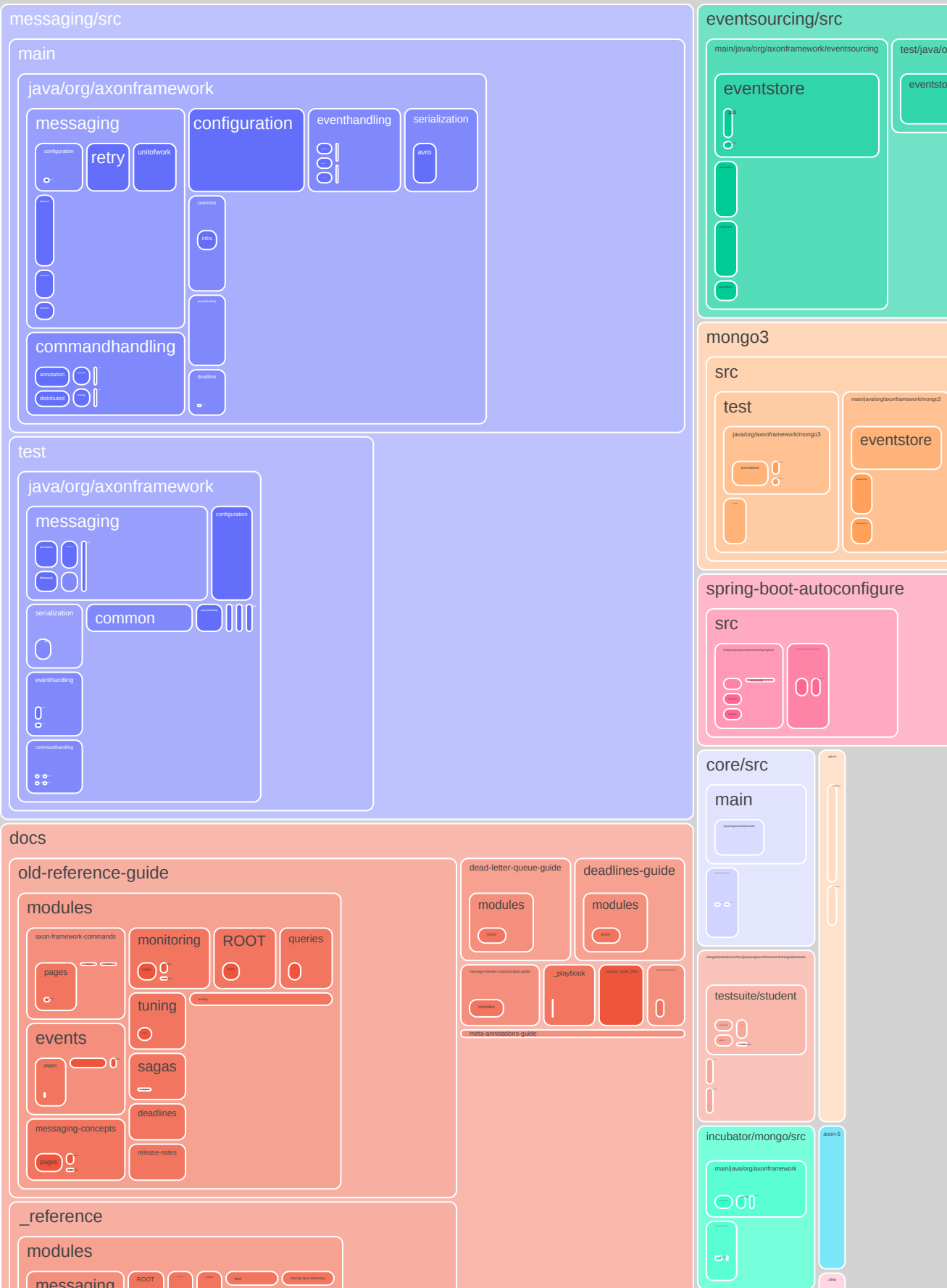
	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	317.000000	317.000000	317.000000	317.000000	317.000000	317.000000
mean	13.738170	7.463722	333.952681	400.574132	640.056782	558.889590
std	57.010194	6.974827	1435.127527	1033.256021	1204.027191	1216.281516
min	1.000000	2.000000	2.000000	1.000000	2.000000	1.000000
25%	1.000000	3.000000	23.000000	5.000000	60.000000	5.000000
50%	3.000000	6.000000	63.000000	49.000000	219.000000	94.000000
75%	8.000000	8.000000	168.000000	257.000000	354.000000	298.000000
max	913.000000	71.000000	22052.000000	4798.000000	4885.000000	4797.000000

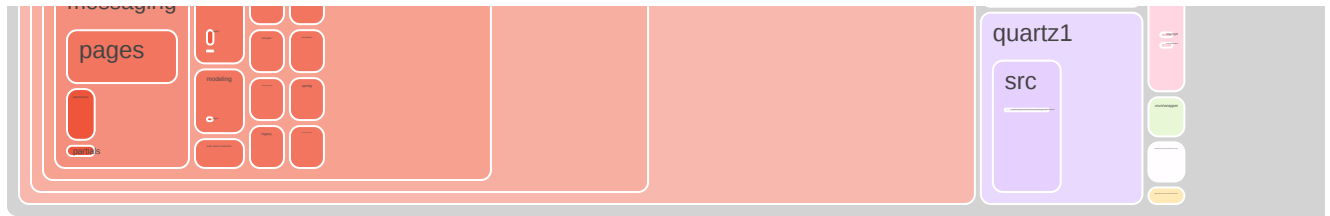
	directoryPath	directoryParentPath
0	AxonFramework-4.11.1/idea/copyright	AxonFramework-4.11.1/idea
1	AxonFramework-4.11.1/idea/inspectionProfiles	AxonFramework-4.11.1/idea
2	AxonFramework-4.11.1/axon-5-module-structure-suggestion/event-sourcing	AxonFramework-4.11.1/axon-5-module-structure-suggestion
3	AxonFramework-4.11.1/axon-5-module-structure-suggestion/integration-tests	AxonFramework-4.11.1/axon-5-module-structure-suggestion
4	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src/test/java/org/axonframework/messaging	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src test/java/org/
5	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging	AxonFramework-4.11.1/axon-5-module-structure-suggestion
6	AxonFramework-4.11.1/axon-5-module-structure-suggestion/test	AxonFramework-4.11.1/axon-5-module-structure-suggestion
7	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/command	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector
8	AxonFramework-4.11.1/core/src/test/java/org/axonframework/commandhandling/disruptor	AxonFramework-4.11.1/core/src/test/java/org/axonframework co
9	AxonFramework-4.11.1/core/src/test/java/org/axonframework/contextsupport/spring	AxonFramework-4.11.1/core/src/test/java/org/axonframework
10	AxonFramework-4.11.1/core/src/test/java/org/axonframework/eventhandling/scheduling/quartz	AxonFramework-4.11.1/core/src/test/java/org/axonframework eventl
11	AxonFramework-4.11.1/docs/_playbook/localLinks	AxonFramework-4.11.1/docs/_playbook
12	AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.1/docs/_reference/modules/ROOT
13	AxonFramework-4.11.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.11.1/docs/_reference/modules/ROOT
14	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
15	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
16	AxonFramework-4.11.1/docs/_reference/modules/configuration/pages	AxonFramework-4.11.1/docs/_reference/modules/configuration
17	AxonFramework-4.11.1/docs/_reference/modules/configuration/partials	AxonFramework-4.11.1/docs/_reference/modules/configuration
18	AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.1/docs/_reference/modules/disruptor
19	AxonFramework-4.11.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.11.1/docs/_reference/modules/disruptor
20	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
21	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
22	AxonFramework-4.11.1/docs/_reference/modules/legacy/pages	AxonFramework-4.11.1/docs/_reference/modules/legacy
23	AxonFramework-4.11.1/docs/_reference/modules/legacy/partials	AxonFramework-4.11.1/docs/_reference/modules/legacy
24	AxonFramework-4.11.1/docs/_reference/modules/metrics/pages	AxonFramework-4.11.1/docs/_reference/modules/metrics
25	AxonFramework-4.11.1/docs/_reference/modules/metrics/partials	AxonFramework-4.11.1/docs/_reference/modules/metrics
26	AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.1/docs/_reference/modules/micrometer
27	AxonFramework-4.11.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.11.1/docs/_reference/modules/micrometer
28	AxonFramework-4.11.1/docs/_reference/modules/modeling/partials	AxonFramework-4.11.1/docs/_reference/modules/modeling
29	AxonFramework-4.11.1/docs/_reference/modules/spring/pages	AxonFramework-4.11.1/docs/_reference/modules/spring

Number of files per directory

Directories and their file count

AxonFramework-4.11.1





Most frequent file extension per directory

AxonFramework-4.11.1

docs

- _reference
 - modules
 - messaging
 - pages
 - axion_messages
 - deadlines
 - partials
 - attachments
 - axon-server-connector
 - pages
 - partials
 - eventsourcing
 - pages
 - partials
 - micrometer
 - pages
 - partials
 - spring
 - pages
 - partials
 - configuration
 - pages
 - partials
 - legacy
 - pages
 - partials
 - springboot-starter
 - pages
 - partials
 - tracing-open-telemetry
 - pages
 - partials
 - ROOT
 - attachments
 - partials
 - pages
 - disruptor
 - pages
 - partials
 - metrics
 - pages
 - partials
 - test
 - pages
 - partials
 - modeling
 - partials
 - pages
- old-reference-guide
 - modules
 - monitoring
 - partials
 - images
 - axon-framework-commands
 - images
 - partials
 - deadlines
 - partials
 - pages
 - release-notes
 - partials
 - pages
 - testing
 - partials
 - pages
 - queries
 - partials
 - pages
 - tuning
 - partials
 - pages
 - messaging-concepts
 - partials
 - images
 - events
 - partials
 - images
 - sagas
 - partials
 - pages
 - ROOT
 - images
 - pages
- dead-letter-queue-guide
 - modules
 - ROOT
 - deadlines-guide
 - modules
 - ROOT
 - meta-annotations-guide
 - modules
 - message-handler-customization-guide
 - modules
 - _playbook
 - localLinks
 - _source_code_links

messaging/src

- main
 - java/org/axonframework
 - commandhandling
 - config
 - gateway
 - retry
 - tracing
- messaging
 - reflection
 - timeout
 - annotation
 - configuration
 - reflection
- test
 - java/org/axonframework
 - messaging
 - configuration
 - timeout
 - annotation
 - commandhandling
 - annotation
 - gateway
 - retry
 - tracing

modelling/src

- main
 - java/org/axonframework/modelling
 - saga
 - repository
 - command
 - configuration
 - annotation
- test/java/org/axonframework/modelling
 - repository
 - saga/repository
 - annotation
 - command
 - configuration

resources/META-INF/services

mongo3

- src
 - test
 - java/org/axonframework/mongo3
 - mongosuite
 - saga/repository
 - serializer/bson
 - eventstore
 - common
 - saga/repository
 - serializer/bson
 - eventstore
- resources

spring-boot-autoconfigure

- src
 - test/java/org/axonframework/springboot
 - fixture/avro
 - test1
 - test2
 - integration
 - main/java/org/axonframework/springboot
 - service/connection
 - autoconfig

incubator/mongo/s

- main/java/org/axonframework
 - eventstore/mongo
 - util/mongo

core/src

- test/java/org/axonframework
 - eventhandling/scheduling/quartz

axon-5-module-structure-suggestion

- messaging-core
 - src
- integration-tests
- messaging
- test

eventsourcing/src

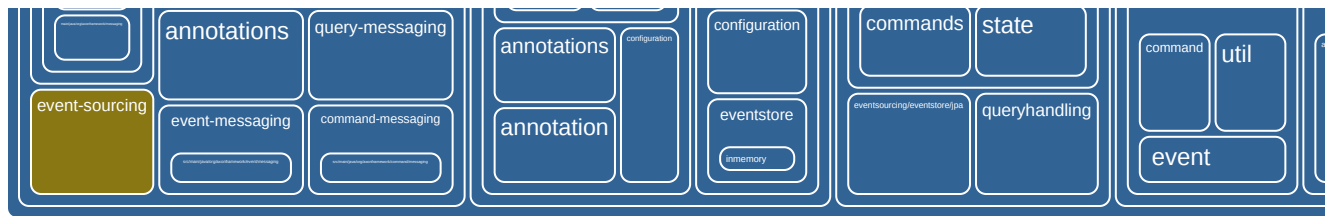
- main/java/org/axonframework/eventsourcing
 - eventstore
 - inmemory
 - jpa
- annotation

testsuite/student

- common
- events

axon-server-connector

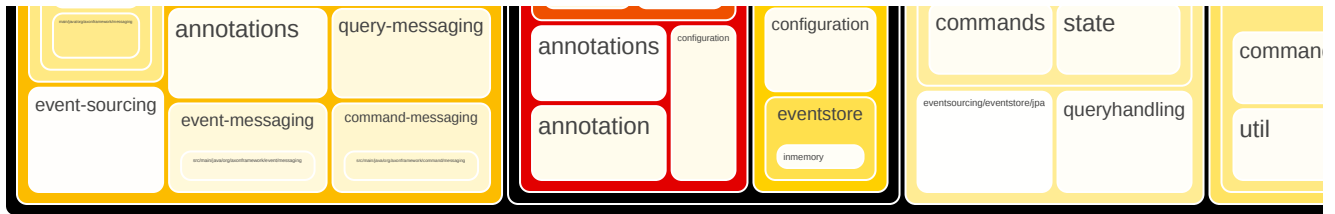
- main/java/org/axonframework/axonserver/connector



Number of commits per directory

Number of git commits





Number of distinct authors per directory

AxonFramework-4.11.1

docs

_reference

modules

messaging

pages

advanced_messaging

messages

deadlines

routing

partials

attachments

axon-server-connector

pages

partials

configuration

pages

partials

legacy

pages

partials

springboot-starter

pages

partials

tracing-open-telemetry

pages

partials

micrometer

pages

partials

spring

pages

partials

ROOT

attachments

partials

pages

disruptor

pages

partials

metrics

pages

partials

test

pages

partials

modeling

partials

pages

dead-letter-queue-guide

modules

ROOT

examples

partials

pages

deadlines-guide

modules

ROOT

partials

pages

identifier-generation-guide

modules

meta-annotations-guide

modules

message-handler-customization-guide

modules

_playbook

localLinks

_source_code_links

old-reference-guide

modules

monitoring

partials

images

pages

axon-framework-commands

images

partials

pages

deadlines

partials

pages

testing

partials

pages

sagas

partials

pages

messaging-concepts

partials

images

pages

events

partials

images

pages

release-notes

partials

pages

queries

partials

pages

tuning

partials

pages

ROOT

images

pages

modelling/src

main

java/org/axonframework/modelling

saga

command

repository

configuration

annotation

resources/META-INF/services

test/java/org/axonframework/modelling

repository

saga/repository

annotation

command

configuration

mongo3

src

test

java/org/axonframework/mongo3

common

saga/repository

mongoutils

serializer/bson

saga/repository

eventstore

serializer/bson

eventstore

criteria

resources

spring-boot-autoconfigure

src

test/java/org/axonframework/springboot

fixture/avro

autoconfig

connection

test1

test2

integration

main/java/org/axonframework/springboot

service/connection

autoconfig

incubator

main/java/org/axonframework/incubator

eventstore/memory

util/memory

core/src

test/java/org/axonframework/core

commandhandling/impl

contextsupport/impl

axon-server

main/java/org/axonframework/axonserver

commandhandling/impl

contextsupport/impl

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inmemory

jpa

annotation

eventsourcing/src

main/java/org/axonframework/eventsourcing

test/java/org/axonframework/eventsourcing

eventstore

inmemory

jpa

annotation

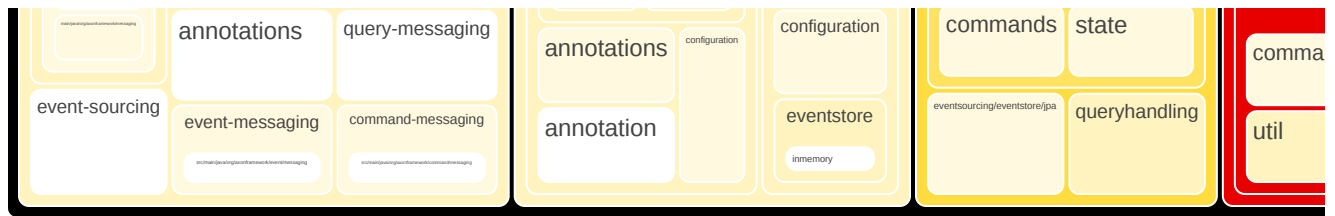
spring-boot-autoconfigure

integrationtests/src/test/java/org/axonframework/integrationtests

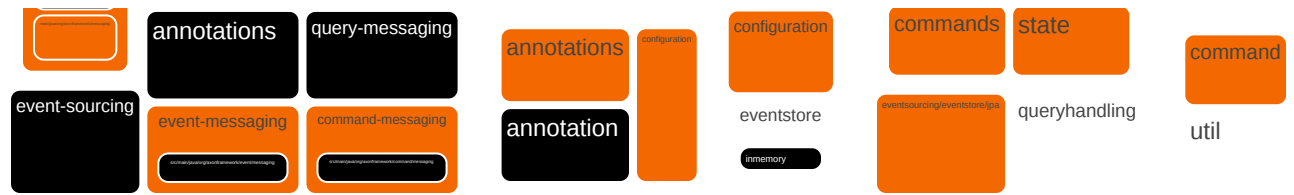
testsuite/student

common

events

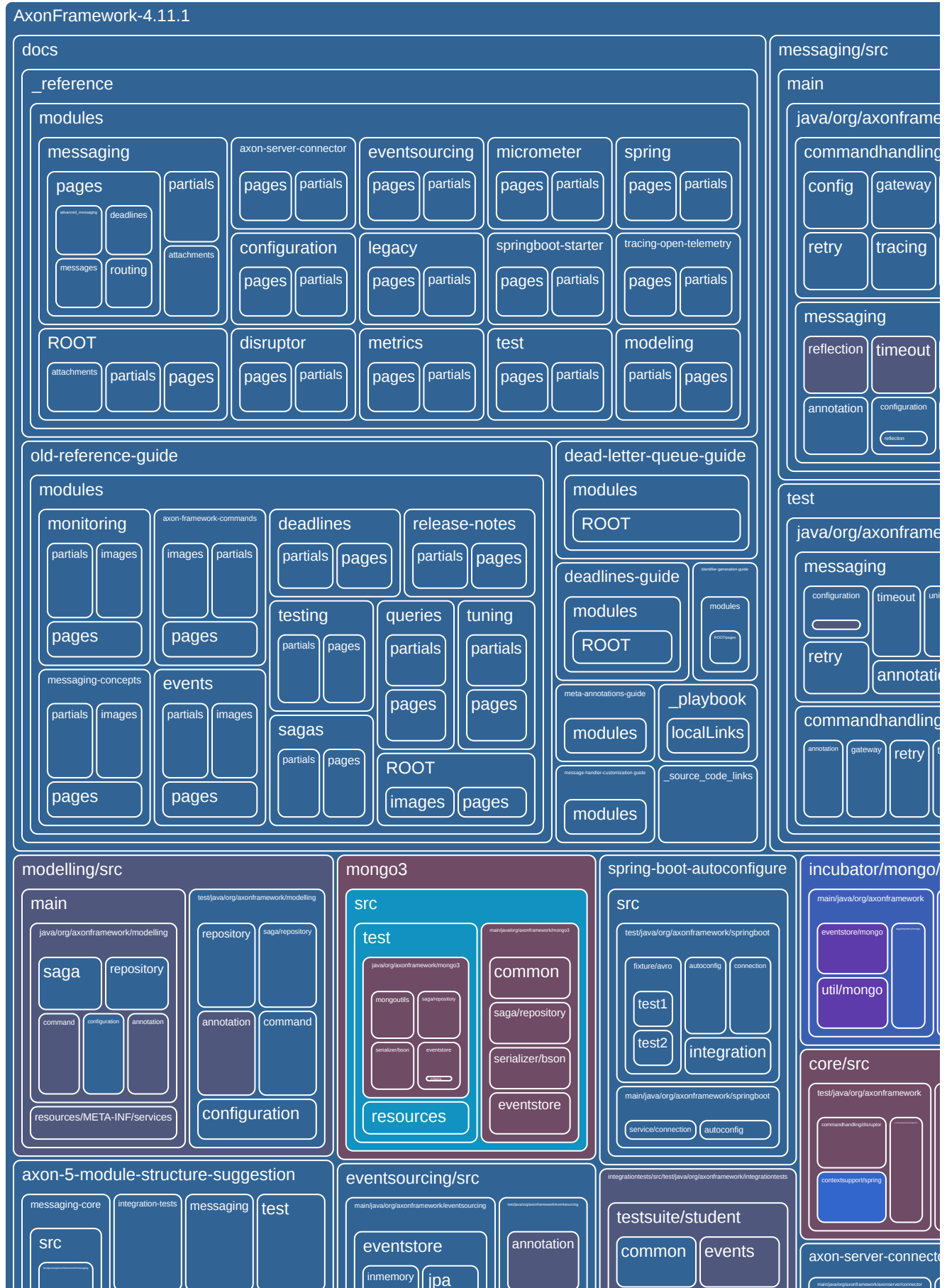


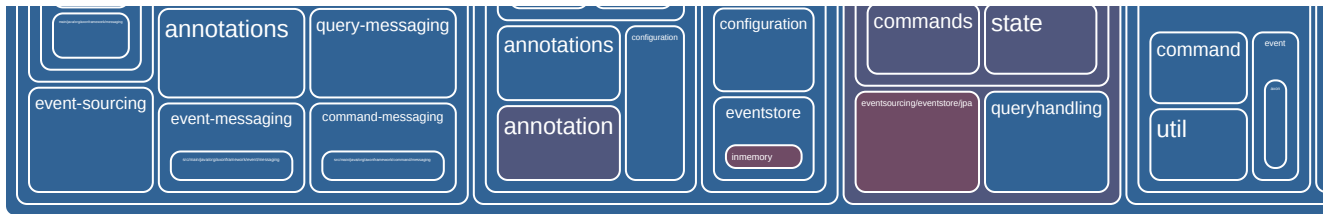
Directories with very few different authors



Main author per directory

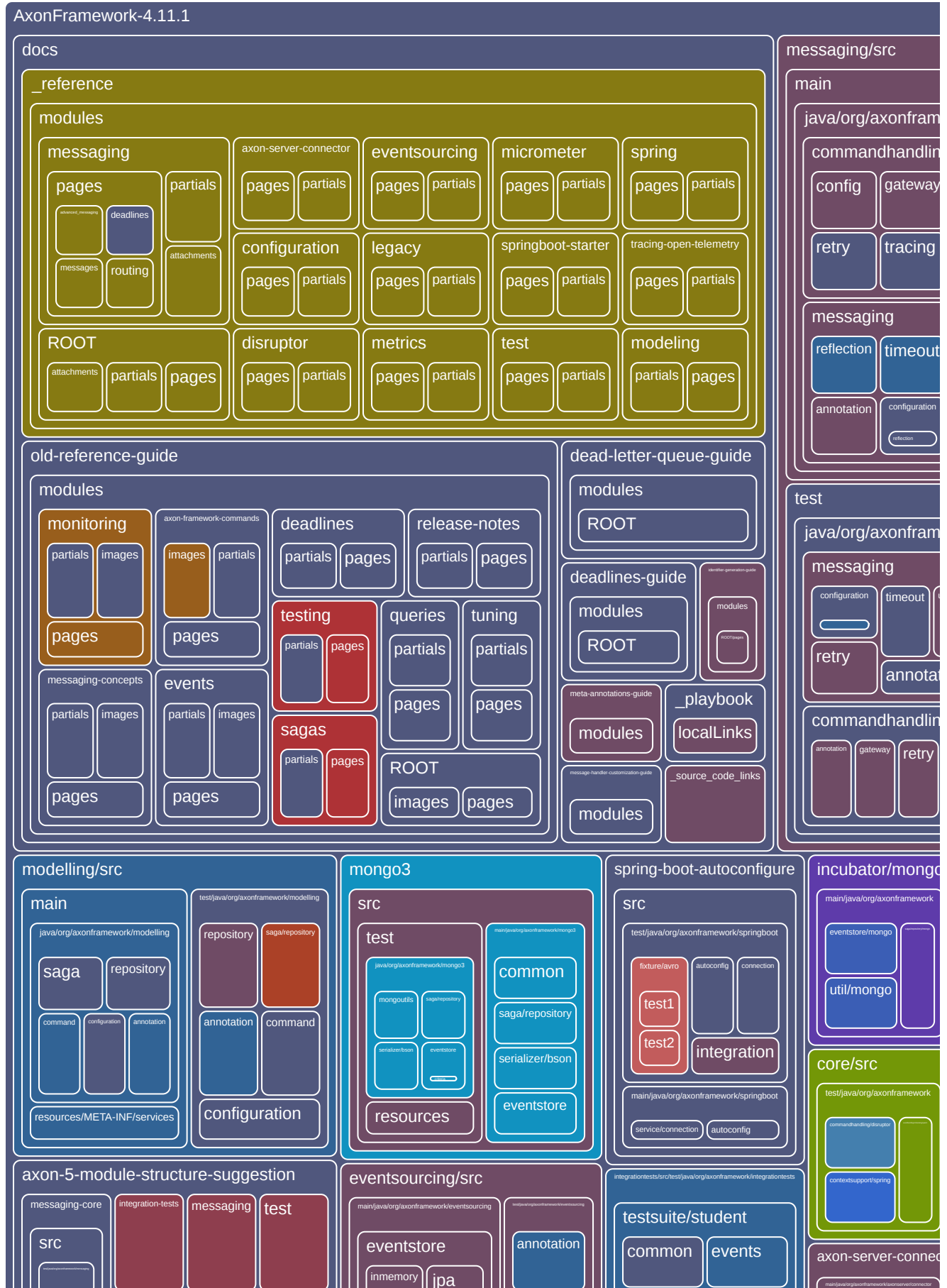
Main authors with highest number of commits

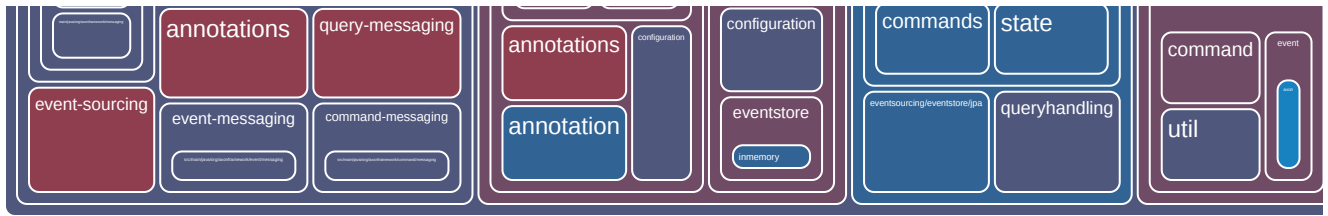




Second author per directory

Second author with the second highest number of commits



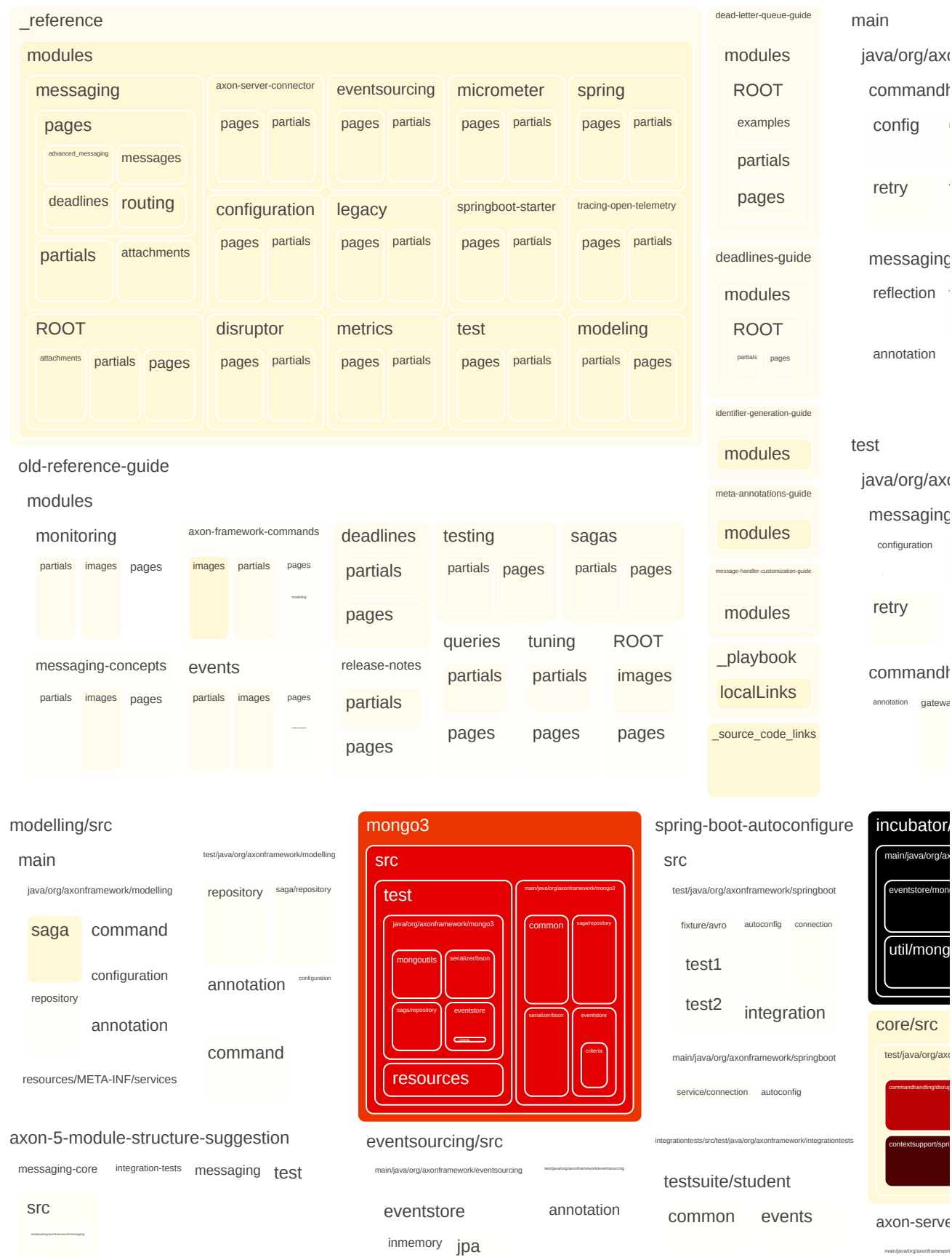


Days since last commit per directory

Days since last commit

AxonFramework-4.11.1

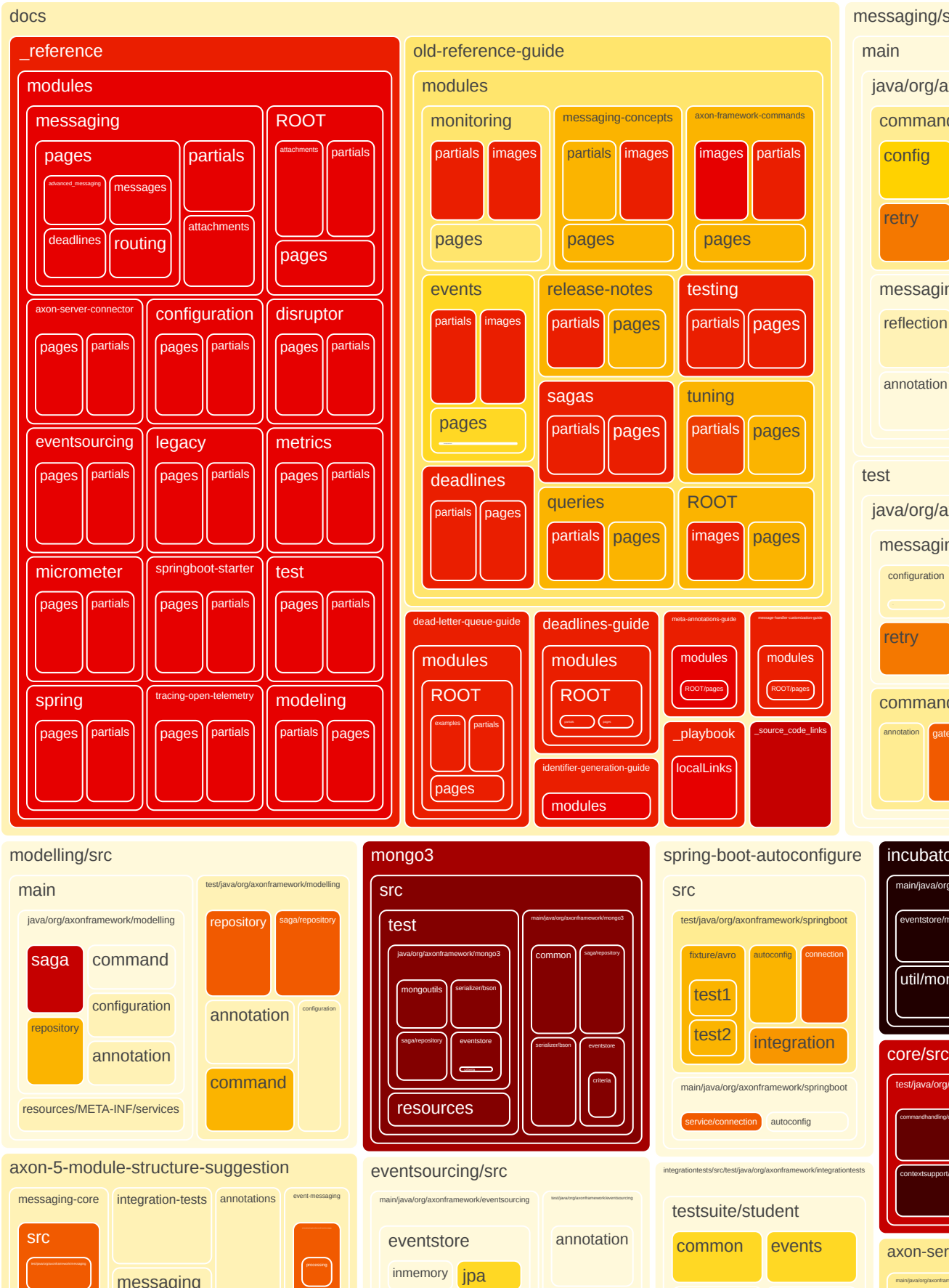
docs

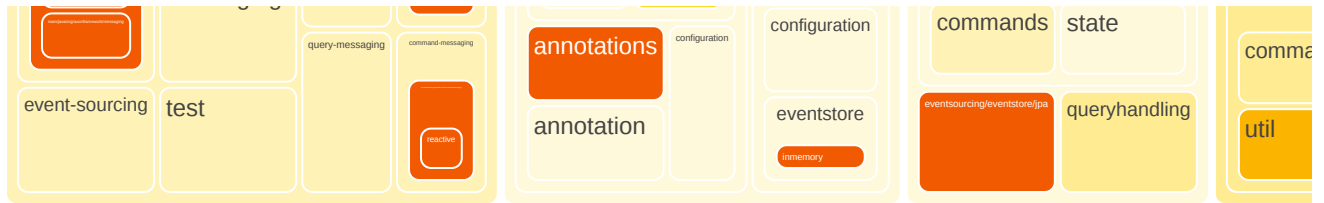




Rank of days since last commit

AxonFramework-4.11.1

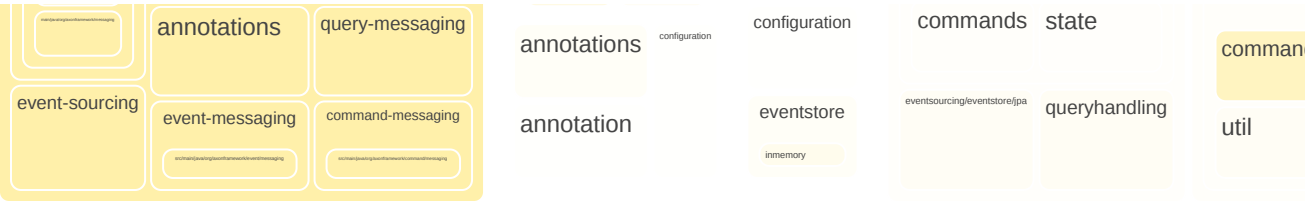




Days since last file creation per directory

messaging/src

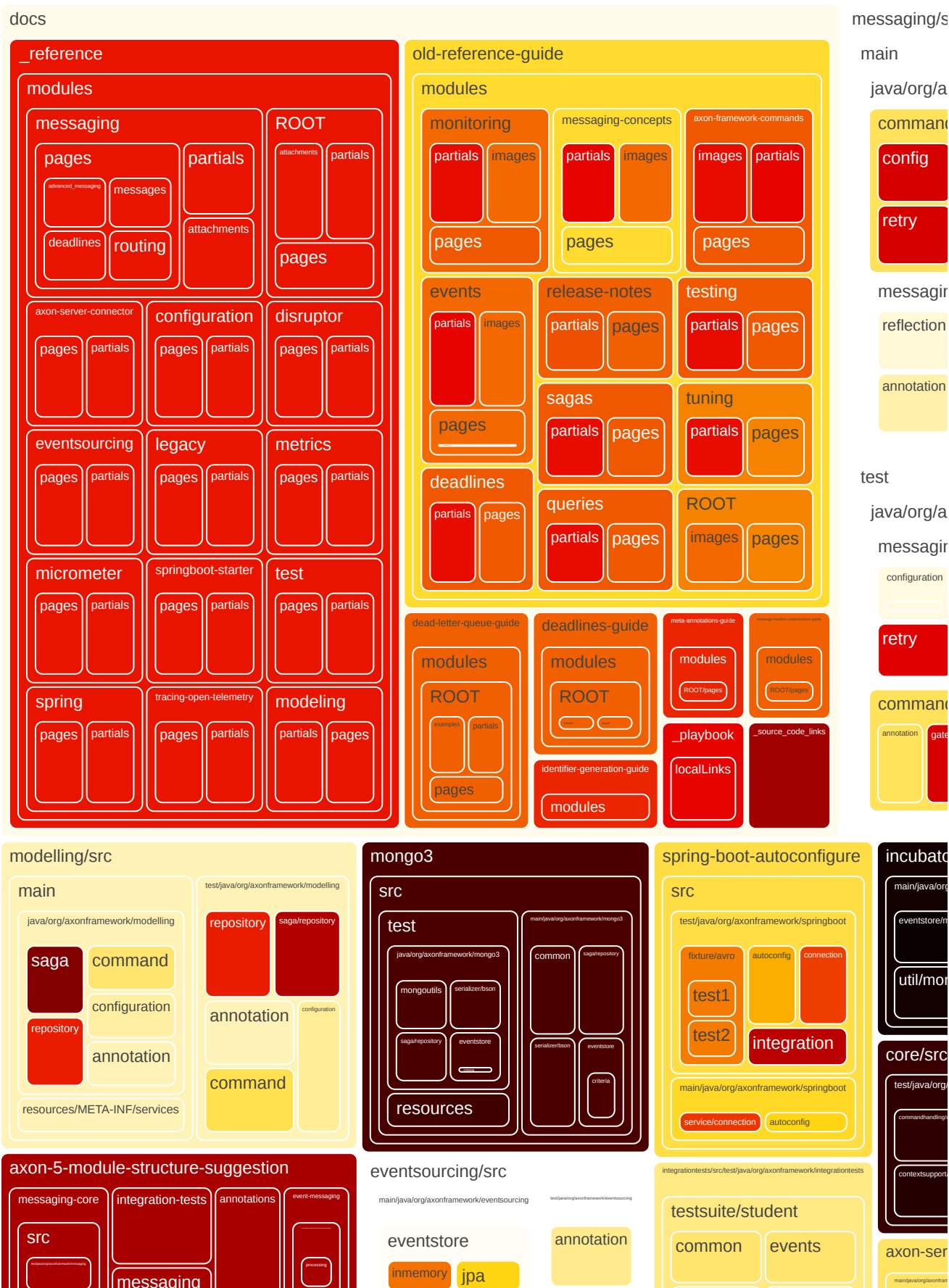
main

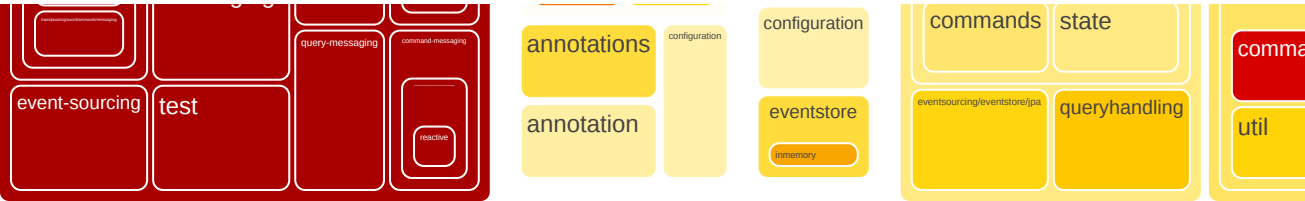


Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.1



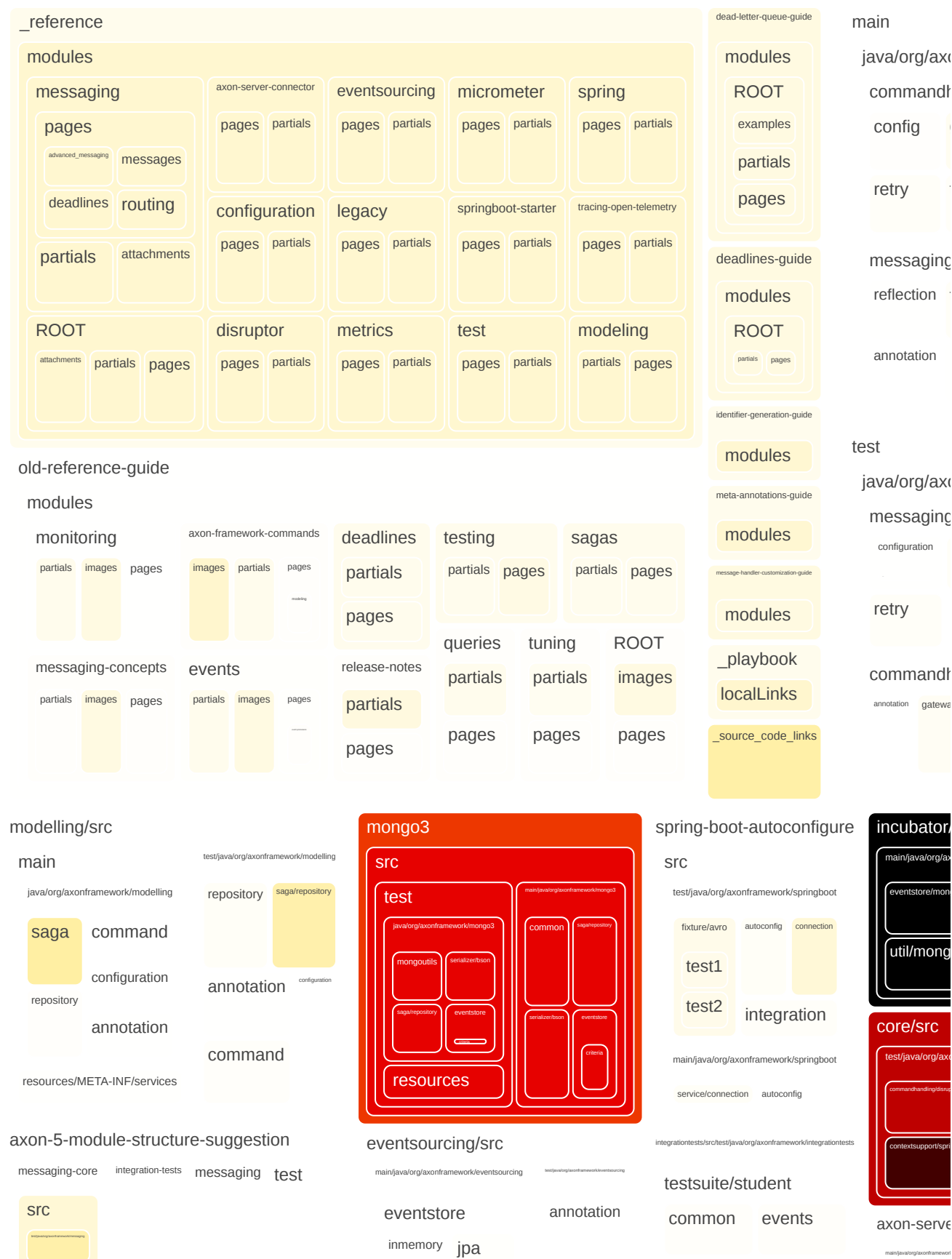


Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.1

docs

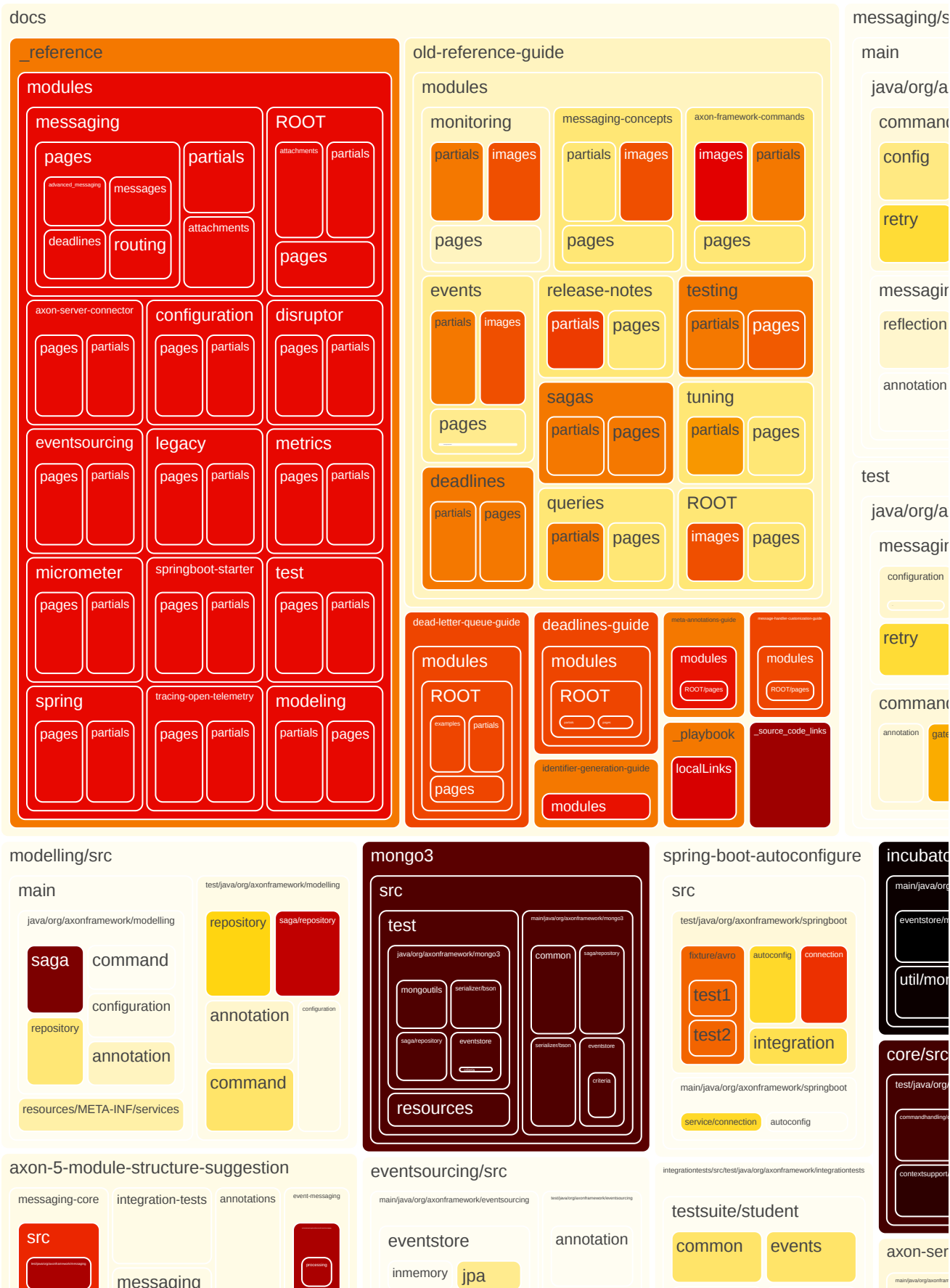


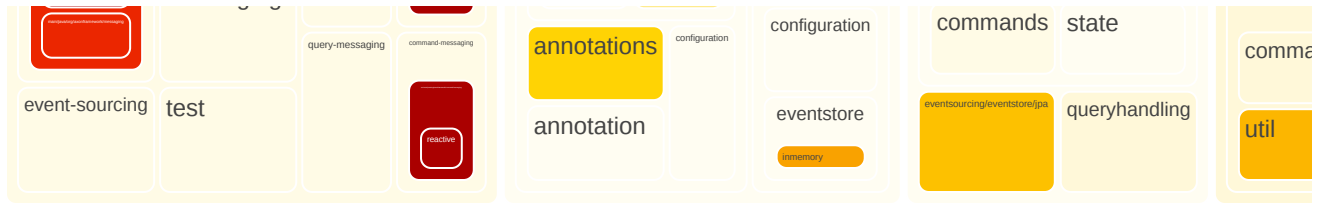


Days since last file modification per directory (ranked)

Rank of days since last file modification

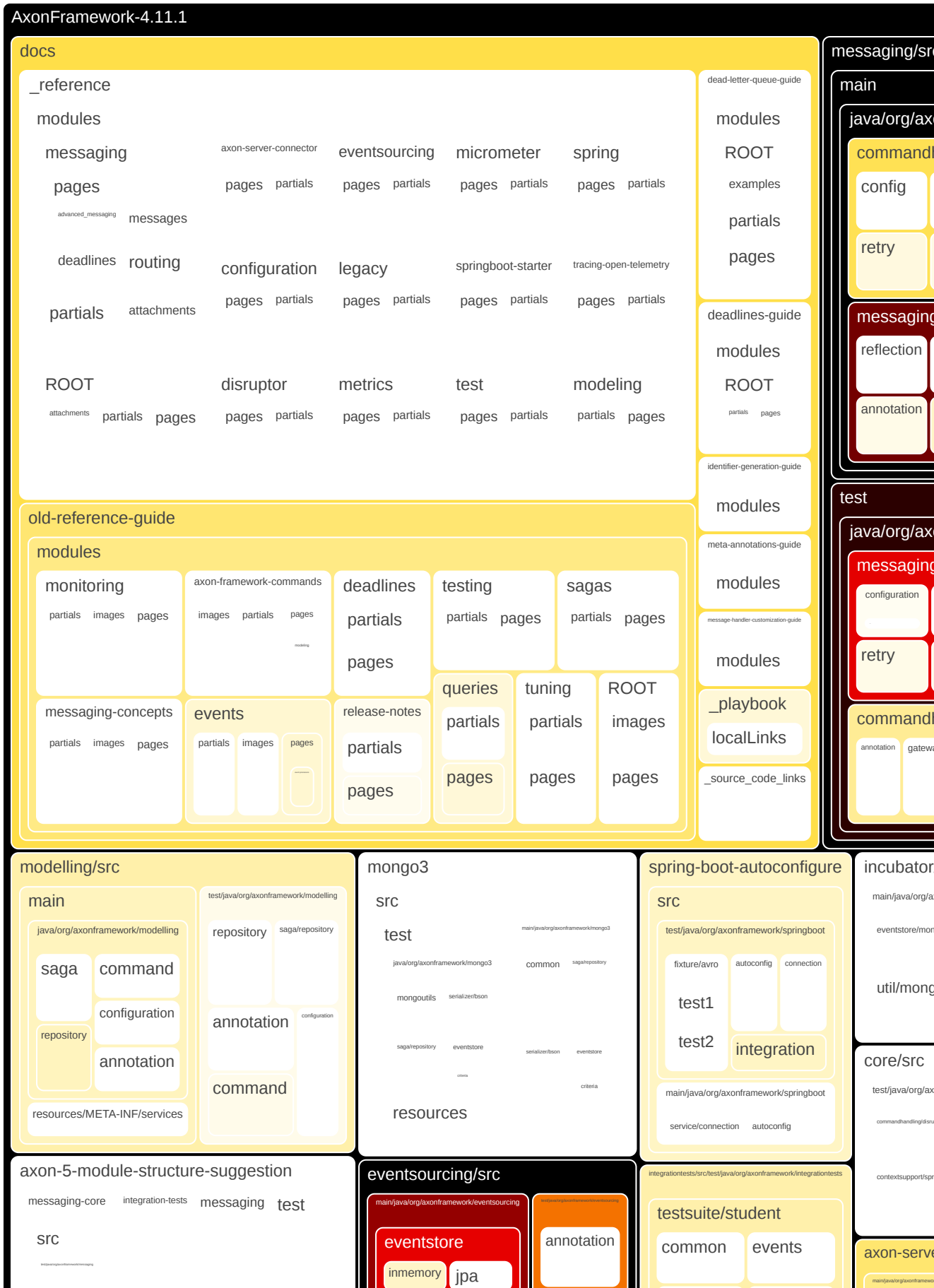
AxonFramework-4.11.1

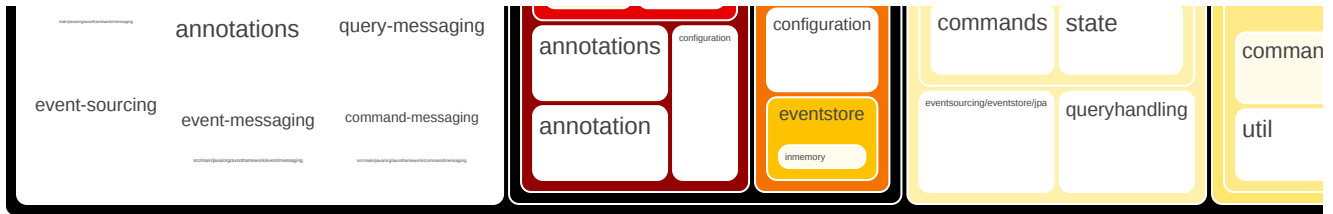




File changed frequently with other files

Pairwise file changes





Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

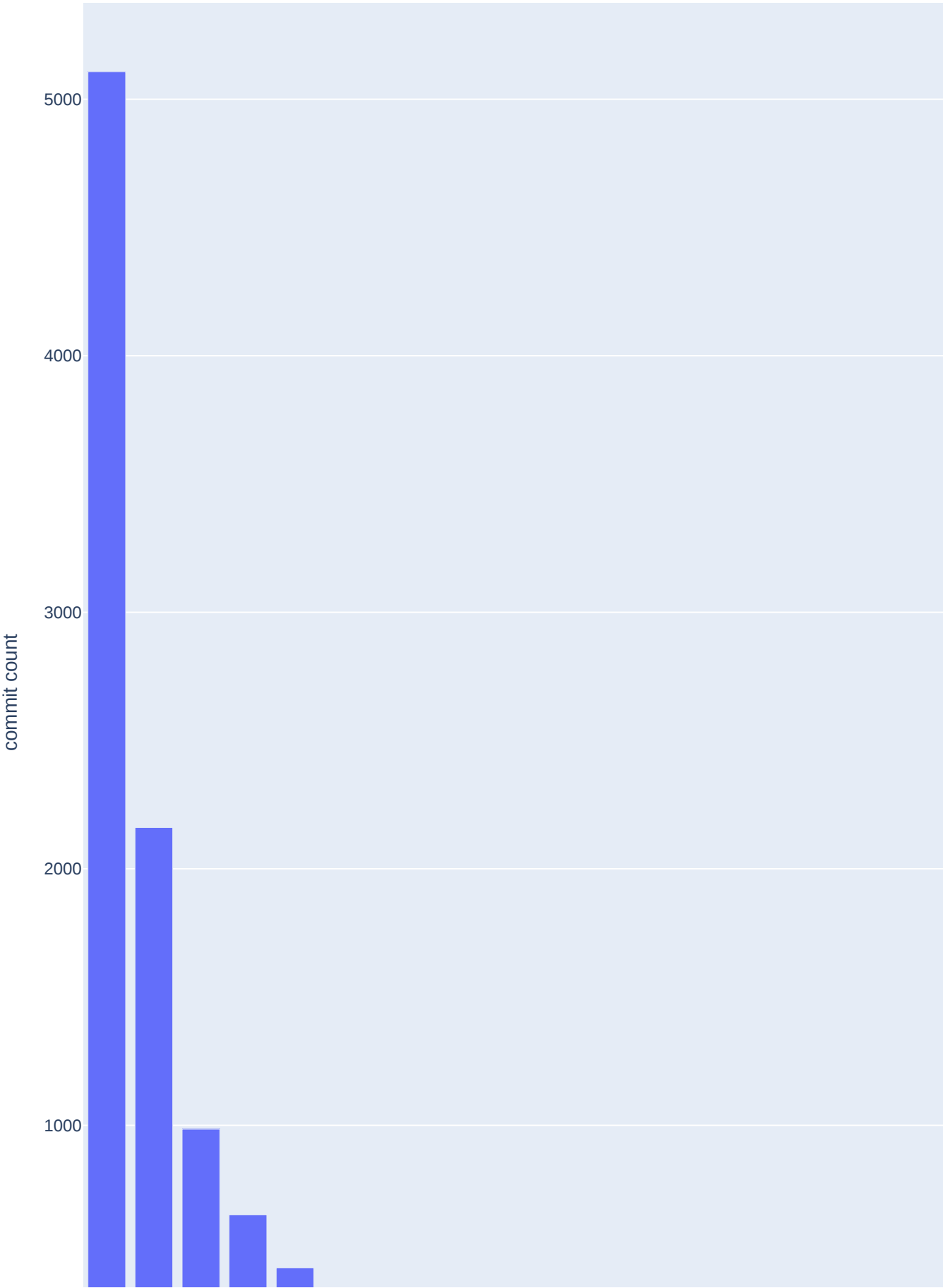
Sum of commits that changed more than 30 files (each) = 1185
Max changed files with one commit = 4041

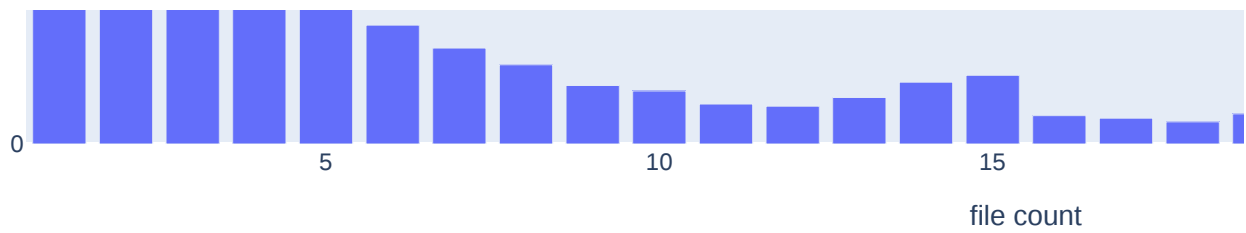
	filesPerCommit	commitCount
count	375.000000	375.00000
mean	399.136000	34.92800
std	560.984963	293.94355
min	1.000000	1.00000
25%	94.500000	1.00000
50%	203.000000	2.00000
75%	430.000000	5.00000
max	4041.000000	5107.00000

	filesPerCommit	commitCount
0	1	5107
1	2	2161
2	3	985
3	4	651
4	5	444
5	6	327
6	7	264
7	8	218
8	9	161
9	10	146
10	11	110
11	12	104
12	13	128
13	14	170
14	15	189
15	16	78
16	17	71
17	18	61
18	19	83
19	20	73
20	21	45
21	22	69
22	23	36
23	24	37
24	25	43
25	26	34
26	27	33
27	28	32
28	29	29
29	30	24

Bar chart with the number of files per commit distribution

Changed files per commit





WordCloud of git authors

	word	frequency
0	Steven van Beelen	4365
1	Allard Buijze	3156
2	smcvb	822
3	Mitchell Herrijgers	457
4	Milan Savic	449
5	Gerard Klijs	246
6	Marc Gathier	231
7	sara pellegrini	163
8	Mateusz Nowak	151
9	Rene de Waele	129

Wordcloud of git authors

